

Lecture 6: CPU Organization & Internal Structure

Learning Objectives

By the end of this lecture, students will be able to:

- Define the CPU (Central Processing Unit).
 - Explain why the CPU is called the "brain" of the computer.
 - Identify the main components inside the CPU.
 - Understand the role of the ALU, Control Unit, Registers, and Cache.
 - Explain how these components work together.
 - Understand the CPU's role in executing a program.
 - Relate CPU organization to Assembly Language.
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1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally mention:

"In today's Computer Organization and Assembly Language lecture, we'll study CPU Organization, including the Control Unit, ALU, Registers, Cache Memory, and how these components work together to execute instructions."

In the previous lectures, we studied the basic components of a computer system and explored the Von Neumann and Harvard architectures.

We learned how instructions and data are stored in memory and how the CPU communicates with them.

But today, we're going to answer an even more interesting question:

What is actually inside the CPU?

We often hear people say that the CPU is the "brain of the computer."

But what makes it so important?

Does the CPU perform every task by itself?

Or does it contain smaller components that work together?

Today's lecture will answer all of these questions.

By the end of this lecture, you'll understand the internal structure of a CPU and how each component contributes to executing a program.

So let's get started.

2. What is a CPU?

Definition

CPU (Central Processing Unit) is the primary component responsible for executing program instructions, processing data, and controlling the overall operation of the computer.

Simply put:

The CPU is the component that executes instructions.

Whenever you:

- Open Chrome
- Run VS Code
- Play a game
- Open Microsoft Word
- Run a Python program

The CPU is responsible for executing the instructions required for these tasks.

3. Why is the CPU Called the Brain of the Computer?

Think about the human body.

Your brain:

- Makes decisions
- Processes information
- Sends signals
- Controls body functions

Similarly, the CPU:

- Processes instructions
- Performs calculations
- Controls hardware operations
- Coordinates communication between components

However, it's important to remember:

The CPU is the processing center, but it is not the only important component.

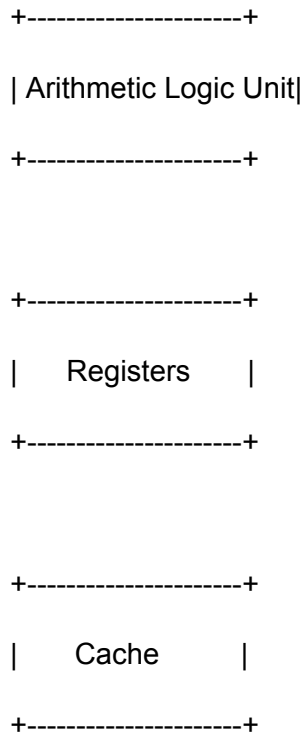
It still depends on:

- Memory
- Storage
- Input devices
- Output devices

4. Block Diagram of CPU Organization

CPU





These four components work together to execute every instruction.

5. Main Components of the CPU

The CPU mainly consists of:

1. Control Unit (CU)
2. Arithmetic Logic Unit (ALU)
3. Registers
4. Cache Memory

We'll briefly introduce each one now and study them in more detail in the next lectures.

6. Arithmetic Logic Unit (ALU)

The **ALU** is responsible for performing:

Arithmetic Operations

- Addition
- Subtraction
- Multiplication
- Division

Logical Operations

- AND
- OR
- NOT
- XOR

Comparison Operations

- Greater than
- Less than
- Equal to

Example

Suppose the program contains:

```
int c = 10 + 20;
```

The ALU performs the addition and produces the result:

10 + 20

↓

30

The ALU does **not** decide *when* to perform the operation—that decision comes from the Control Unit.

7. Control Unit (CU)

The **Control Unit** is responsible for coordinating all CPU operations.

It:

- Fetches instructions
- Decodes instructions
- Sends control signals
- Coordinates data movement
- Directs the ALU
- Controls memory access

Think of the Control Unit as the **manager** of the CPU.

Analogy

Imagine a classroom.

- The **teacher** gives instructions.
- The **students** perform the work.

Similarly:

- The **Control Unit** gives instructions.
 - The **ALU**, **Registers**, and **Memory** perform their respective tasks.
-

8. Registers

Registers are:

Small, high-speed storage locations inside the CPU.

They temporarily store:

- Data
- Instructions
- Memory addresses
- Intermediate results

Example:

Register A = 15

Register B = 25

The ALU uses these values directly:

$15 + 25$

↓

40

Registers are much faster than RAM because they are located inside the CPU.

9. Cache Memory

Between the CPU and RAM is **Cache Memory**.

CPU

↓

Cache

↓

RAM

Cache stores frequently used instructions and data.

This reduces the time needed to access RAM.

Why is Cache Needed?

Suppose the CPU needs the same value many times.

Without cache:

CPU

↓

RAM

↓

CPU

↓

RAM

↓

CPU

The CPU repeatedly waits for RAM.

With cache:

CPU

↓

Cache

↓

CPU

↓

Cache

Access is much faster.

10. Complete CPU Structure

CPU

+-----+

| Control Unit |

+-----+

+-----+

| ALU |

+-----+

+-----+

| Registers |

+-----+

+-----+

| Cache Memory |

+-----+

11. How the CPU Executes a Program

Let's consider this C++ code:

```
int x = 5;
```

```
int y = 7;
```

```
int z = x + y;
```

Step 1

The program is loaded into RAM.

↓

Step 2

The Control Unit fetches the first instruction.

↓

Step 3

The instruction is decoded.

↓

Step 4

The values 5 and 7 are loaded into registers.

↓

Step 5

The ALU performs the addition.

↓

Step 6

The result (12) is stored in a register and then written back to memory.

12. CPU Working Diagram

Program



Memory



Control Unit



Registers



ALU



Registers



Memory

↓

Output

13. CPU and Assembly Language

This is where Assembly Language becomes important.

In high-level languages, you simply write:

```
a = b + c;
```

The compiler converts this into assembly instructions.

For example, in MIPS (conceptually):

```
LW $t0, A
```

```
LW $t1, B
```

```
ADD $t2, $t0, $t1
```

```
SW $t2, C
```

Explain that:

- **LW** loads data from memory into a register.
- **ADD** performs the addition in the ALU.
- **SW** stores the result back into memory.

Tell students that they don't need to memorize these instructions yet; they'll learn MIPS programming later in the course.

14. CPU Communication

The CPU constantly communicates with:

CPU

↓

Memory

↓

Input Devices

↓

Output Devices

For example:

- Keyboard → Input
- CPU processes data
- Monitor → Output

15. CPU Clock

The CPU performs operations according to a **clock**.

The clock generates regular timing signals.

Each clock cycle allows the CPU to perform part of an instruction.

Example:

Clock

↓

Tick

↓

Tick

↓

Tick

↓

Tick

Modern CPUs perform billions of clock cycles per second.

A processor running at **3 GHz** has a clock frequency of approximately **3 billion cycles per second**.

16. CPU Performance Depends On

CPU performance is influenced by several factors, including:

- Clock speed
- Number of cores
- Cache size
- Instruction Set Architecture (ISA)
- Microarchitecture
- Memory subsystem

It's not determined by clock speed alone.

17. Real-Life Analogy

Imagine a restaurant.

Customer

Places an order.

↓

Manager (Control Unit)

Receives the order.

↓

Chef (ALU)

Prepares the food.

↓

Table (Registers)

Temporarily holds the prepared food.

↓

Storage Room (Memory)

Stores ingredients and supplies.

The restaurant works efficiently because each part has a specific responsibility.

The CPU operates in a similar coordinated way.



Lecture Summary

Today, we explored the internal organization of the CPU and learned why it is considered the brain of the computer.

We studied the four main components inside the CPU: the **Control Unit**, **Arithmetic Logic Unit (ALU)**, **Registers**, and **Cache Memory**.

We also learned how these components work together to execute a program, from fetching instructions to performing calculations and storing results.

Finally, we introduced the role of the CPU clock and discussed some of the factors that affect CPU performance.

In the next lecture, we'll take a deeper look at the **Arithmetic Logic Unit (ALU)** and understand how it performs arithmetic operations, logical operations, comparisons, and works with status flags.

Thank you for watching, and I'll see you in the next lecture.
